Exercises Sheet 3

1 (a) $X_t = Z_t + \frac{1}{4} Z_{t-1} - \frac{1}{8} Z_{t-2}$

MA(2) process

Invertible if...?

$$X_t = \left(1 + \frac{B}{4} - \frac{B^2}{8}\right) Z_t$$
$$= \theta(B) Z_t$$

Invertible if $\theta(z) = 0$ has roots outside the unit circle.

Set $1 + \frac{z}{4} - \frac{z^2}{8} = 0$

$$\left(1 + \frac{z}{2}\right)\left(1 - \frac{z}{4}\right) = 0$$

Roots : $z = -2$ or 4

Each is s.t. $|z| > 1 \Rightarrow$ process is invertible.

(b) The AGF is given by

$$\Gamma(z) = \sigma_Z^2 \, \theta(z)\, \theta(z^{-1})$$

where $X_t = \theta(B) Z_t$.

Here

$$\Gamma(z) = \sigma_Z^2 \left[\left(1 + \frac{z}{4} - \frac{z^2}{8}\right)\left(1 + \frac{1}{4z} - \frac{1}{8z^2}\right) \right]$$

$$= \sigma_Z^2 \left[-\frac{1}{8z^2} + \frac{7}{32z} + \frac{69}{64} + \frac{7}{32} z - \frac{1}{8} z^2 \right]$$

(c) 「Coefficient of z^h is $\gamma(h)$」

$$\gamma(h) = \begin{cases} \frac{69}{64}\sigma_Z^2 & h = 0; \\ \frac{7}{32}\sigma_Z^2 & |h| = 1; \\ -\frac{1}{8}\sigma_Z^2 & |h| = 2; \\ 0 & |h| \geq 3. \end{cases}$$

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)}.$$

3. If we assume that, approximately, r_j is normally distributed then, if the ρ_j value is zero then we should expect r_j to lie in the interval $\left(-\frac{2}{\sqrt{n}}, \frac{2}{\sqrt{n}}\right)$

Here $n = 100 \Rightarrow \left(-\frac{2}{10}, \frac{2}{10}\right) = (-0.2, 0.2)$

We have $|r_1|, |r_2|$ and $|r_5|$ are all greater than 0.2.

$|r_1|$ and $|r_2|$ are similar in magnitude and the sample ACF does not appear to decline gradually or exponentially to zero.

Guess that an MA(2) process would be suitable.

4. The true process is

$$X_t = (1 - \theta B) Z_t \qquad (MA(1)).$$

The fitted process is:

$$(1 - \phi B) X_t = Z'_t$$

$\{Z'_t\}$ is the set of residuals from the fitted process.

Here

$$\begin{aligned} Z'_t &= (1 - \phi B) X_t \\ &= (1 - \phi B)(1 - \theta B) Z_t \qquad \text{(from (1))} \\ &= (1 - \phi B - \theta B + \phi\theta B^2) Z_t \\ &= Z_t - (\phi + \theta) Z_{t-1} + \phi\theta Z_{t-2} \end{aligned}$$

The residuals $\{Z'_t\}$ are an MA(2) process.

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5. Let $\{Z'_t\}$ be the residuals from the fitted model.

$\{Z'_t\}$ is an AR(1) process with

$$(1 - 0.4B) Z'_t = Z_t$$

where $\{Z_t\}$ is a white noise process.

$$\Rightarrow \quad Z'_t = (1 - 0.4B)^{-1} Z_t \qquad (*)$$

The model that was fitted is:

$$X_t = (1 - 0.6B) Z'_t$$

$\therefore$ From $(*)$, we have

$$X_t = (1 - 0.6B)(1 - 0.4B)^{-1} Z_t$$

$$\Rightarrow (1 - 0.4B) X_t = (1 - 0.6B) Z_t$$

$$\Rightarrow X_t - 0.4 X_{t-1} = Z_t - 0.6 Z_{t-1}$$

We should fit an ARMA(1,1) model with $\phi_1 = -0.4$; $\theta_1 = -0.6$.

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6. Given $\bar{X}_n = \frac{1}{n} \sum_{t=1}^{n} X_t$

$$\Rightarrow \text{Var}(\bar{X}_n) = \text{Var}\left(\frac{1}{n} \sum_{t=1}^{n} X_t\right)$$

$$= \frac{1}{n^2} \text{Var}\left(\sum_{t=1}^{n} X_t\right)$$

$$= \frac{1}{n^2} \left[\sum_{t=1}^{n} \text{Var}(X_t) + \sum_{s \neq t} \text{Cov}(X_t, X_s) \right]$$

$$\begin{pmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \cdots & \text{Cov}(X_1, X_n) \\ & \text{Var}(X_2) & & \\ & & \ddots & \\ \text{Cov}(X_1, X_n) & & & \text{Var}(X_n) \end{pmatrix}$$

$$\operatorname{Var}(\bar{X}_n) = \frac{1}{n^2}\left[n\gamma_X(0) + 2\sum_{r=1}^{n-1}\sum_{t=1}^{n-r} \operatorname{cov}(X_t, X_{t-r})\right]$$

$$= \frac{1}{n^2}\left[n\gamma_X(0) + 2\sum_{r=1}^{n-1}\sum_{t=1}^{n-r} \gamma_X(r)\right]$$

$$= \frac{1}{n^2}\left[n\gamma_X(0) + 2\sum_{r=1}^{n-1}(n-r)\gamma_X(r)\right].$$

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7. $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t$

For $p = 2$ the Yule-Walker equations are given by

$$\begin{pmatrix} 1 & \rho_1 \\ \rho_1 & 1 \end{pmatrix}\begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix} = \begin{pmatrix} \rho_1 \\ \rho_2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix} = \begin{pmatrix} 1 & \rho_1 \\ \rho_1 & 1 \end{pmatrix}^{-1}\begin{pmatrix} \rho_1 \\ \rho_2 \end{pmatrix}$$

$$= \frac{1}{1-\rho_1^2}\begin{pmatrix} 1 & -\rho_1 \\ -\rho_1 & 1 \end{pmatrix}\begin{pmatrix} \rho_1 \\ \rho_2 \end{pmatrix}$$

$$= \frac{1}{1-\rho_1^2}\begin{pmatrix} \rho_1(1-\rho_2) \\ \rho_2 - \rho_1^2 \end{pmatrix}$$

Using the 'hint':

$$(1 - \rho_2^T \underline{\phi}) P_2^{-1} = \left(1 - (\rho_1\ \rho_2)\begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix}\right) P_2^{-1}$$

$$= (1 - \rho_1\phi_1 - \rho_2\phi_2)(P_2^{-1})$$

$$P_2^{-1} = \frac{1}{1-\rho_1^2}\begin{pmatrix} 1 & -\rho_1 \\ -\rho_1 & 1 \end{pmatrix}$$

$$\therefore (1 - \rho_2^T \underline{\phi}) P_2^{-1} = \frac{(1-\rho_1\phi_1 - \rho_2\phi_2)}{1-\rho_1^2}\begin{pmatrix} 1 & -\rho_1 \\ -\rho_1 & 1 \end{pmatrix} \quad (+)$$

$Var(n^{1/2}(\hat{\phi}_j - \phi_j))$ is j^{th} diagonal element of $(+)$.

$$\therefore Var(n^{1/2}(\hat{\phi}_j - \phi_j)) = \frac{1-\rho_1\phi_1 - \rho_2\phi_2}{1-\rho_1^2} \quad (++)$$

Substitute $\begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix} = \frac{1}{1-\rho_1^2}\begin{pmatrix} \rho_1(1-\rho_2) \\ \rho_2 - \rho_1^2 \end{pmatrix}$

into $(++)$.

$$Var(n^{1/2}(\hat{\phi}_j - \phi_j)) = 1 - \phi_2^2.$$